Finite total dephasing and an isolated natural occupation in an open XX chain

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Abstract

We study the stationary one-particle covariance of an open XX spin chain with unequal end reservoirs, a central dephasing defect, and weak dephasing distributed over the chain. The background is scaled so that its total phase-flip event rate stays finite as the chain grows. For fixed specified boundary and defect strengths, an exactly half-occupied natural orbital remains separated from all other natural occupations. We obtain its limiting occupation spacing as an explicit positive function of the total noise strength, while its inverse participation ratio approaches a noise-independent value. Center-deletion interlacing and a two-dimensional sine trial space control the occupation spacing; an orbital residual estimate controls convergence of the normalized mode. The estimates address a spatially nonuniform limit in which the local background vanishes but the integrated resistance changes by a finite amount. The known zero-background exponential profile survives this limit. All limits are taken after stationarity at each finite size, and concern covariance occupations rather than relaxation rates.

Keywords: Quantum Dissipative Systems; Spin chains, ladders and planes

1 Introduction

Boundary-driven XX chains provide explicit examples of nonequilibrium transport modified by dephasing. The uniform-dephasing stationary magnetization and current are known exactly at arbitrary reservoir bias [1]. Related harmonic chains admit a tridiagonal stationary covariance and an additive dephasing resistance [2]. These moment structures make spectral questions about the covariance accessible, even when the full stationary density operator is more complicated.

The eigenvalues of an equal-time one-particle covariance are natural occupations. They differ from the spectrum of the generator that evolves the correlator, as studied, for example, by Alba [3]. Closed two-point dynamics also do not imply a Gaussian many-body state; local dephasing can generate non-Gaussian correlations [4]. Thus spectral isolation of one natural occupation has a precise one-particle meaning, without implying a many-body relaxation gap or an entanglement property.

Here the background noise has a finite total event rate, distributed uniformly over an increasing number of sites. The noise at each fixed site then vanishes, whereas its contribution to the stationary resistance remains finite. This distinction raises a question that a current formula alone does not settle: can a localized natural orbital retain both its shape and a nonzero separation from the other occupations? We answer it on one specified parameter slice, with a central defect in odd open chains. The zero-background geometric profile is already related by site phases to the familiar imaginary-impurity mode [5]. The contribution here is its joint orbital and occupation-spacing limit under finite distributed noise, together with finite-size bounds. Neither the resistance law nor the exponential profile is a new ingredient.

2 Spin model and the finite noise rate

Consider $N = 2L + 1 \geq 3$ spin-1/2 sites on an open path. In units with $\hbar = 1$, the Hamiltonian and canonical dissipator are

$$H = J \sum_{j=1}^{N-1} (\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y), \qquad J > 0, \tag{1}$$

$$\frac{d\rho}{ds} = -\mathrm{i}[H, \rho] + \sum_a \mathcal{D}[L_a]\rho, \qquad \mathcal{D}[A]\rho = A\rho A^\dagger - \tfrac{1}{2}\{A^\dagger A, \rho\}. \tag{2}$$

Here s is physical time. With $\sigma^\pm = (\sigma^x \pm \mathrm{i}\sigma^y)/2$, the end jumps are

$$L_b^+ = \sqrt{\kappa f_b}\, \sigma_b^+, \qquad L_b^- = \sqrt{\kappa(1 - f_b)}\, \sigma_b^-, \qquad b = 1, N, \tag{3}$$

where $f_1 = (1 - \mu)/2$ and $f_N = (1 + \mu)/2$. These are target local occupations of Markovian spin reservoirs. The dephasing jumps are

$$L_j^z = \sqrt{\gamma_j/2}\, \sigma_j^z, \qquad \gamma_j = \gamma_0 + \eta \delta_{j,L+1}. \tag{4}$$

All rates and J have units of inverse time; jump amplitudes have units of inverse square-root time. We fix throughout

$$\kappa = 4J, \qquad \mu = \tfrac{1}{2}, \qquad \eta = 5J, \qquad \gamma_0 = Jq_N, \tag{5}$$

and consider, for each fixed finite dimensionless $c > 0$, either

$$q_N = \frac{c}{N} \quad \text{or} \quad q_N = \frac{c}{N - 1}. \tag{6}$$

Equivalently, one may measure time in J^{-1} and write $J = 1$, $\kappa = 4$, $\eta = 5$. In a convention using a dimensional noise parameter c_{phys}, the c in this article is c_{phys}/J.

The first scaling has a direct stochastic realization. A Poisson clock of rate $Jc/2$ chooses one of the N sites uniformly and applies its Pauli Z operator. Averaging these events produces

$$\mathcal{L}_{\mathrm{bg}}\rho = \frac{Jc}{2N} \sum_{j=1}^{N} (\sigma_j^z \rho \sigma_j^z - \rho), \tag{7}$$

which is exactly the background in Eq. (4). The defect contributes a separate local clock of rate $5J/2$. For $q_N = c/(N-1)$ the background event rate is $JcN/[2(N-1)]$, while the bond-weighted background damping $2(N-1)\gamma_0$ is exactly $2Jc$. Both event rates tend to $Jc/2$. This realization is defined separately for every finite chain; it does not require a uniform choice from an infinite set of sites.

Define Jordan–Wigner operators and the covariance by

$$c_j = \left[\prod_{k<j} (-\sigma_k^z)\right] \sigma_j^-, \qquad G_{jk}(s) = \mathrm{Tr}(\rho(s) c_k^\dagger c_j). \tag{8}$$

In particular $G_{jj} = (1 + \langle \sigma_j^z \rangle)/2$. We first take $s \to \infty$ at fixed N and denote the limit by G_N. Its eigenvalues ν_a lie in $[0, 1]$. The size limit below is always taken after this stationary limit.

3 Exact stationary reduction

The spin adjoint generator closes on the bilinears in Eq. (8):

$$\dot{G}_{jk} = \Big[-\mathrm{i}(hG - Gh) - \tfrac{1}{2}\{K, G\} + P\Big]_{jk} - (\gamma_j + \gamma_k)(1 - \delta_{jk})G_{jk}, \tag{9}$$

where $h_{j,j+1} = h_{j+1,j} = 2J$, $K = \mathrm{diag}(\kappa, 0, \ldots, 0, \kappa)$, and $P = \mathrm{diag}(\kappa f_1, 0, \ldots, 0, \kappa f_N)$. Conjugating a bilinear by σ^z_ℓ changes its sign precisely when $j \neq k$ and ℓ is one of its two endpoints. This yields the last term, which damps off-diagonal entries and leaves densities unchanged directly. The end-spin dissipators give the displayed K and P terms; the Jordan–Wigner string is retained in making this spin calculation. There is no replacement of the right spin bath by an assumed full-state fermion bath.

For completeness, write $v = 2J$, $k_j = K_{jj}$, and

$$d_j = \gamma_j + \gamma_{j+1} + \tfrac{1}{2}(k_j + k_{j+1}), \qquad \mathcal{R} = \frac{4v^2}{\kappa} + \sum_{j=1}^{N-1} d_j, \qquad a = \frac{\mu v}{\mathcal{R}}. \tag{10}$$

The stationary solution has only a diagonal and adjacent coherences:

$$(G_N)_{j,j+1} = -\mathrm{i}a, \qquad (G_N)_{jj} = f_1 + \frac{2va}{\kappa} + \frac{a}{v}\sum_{\ell<j} d_\ell, \tag{11}$$

$$(G_N)_{jk} = 0 \qquad \text{for } |j - k| > 1.$$

The first off-diagonal equation reduces to $v[(G_N)_{j+1,j+1} - (G_N)_{jj}] = ad_j$. The diagonal equations express constant bond current, the second off-diagonal terms cancel because a is constant, and the right boundary equation fixes $\mathcal{R}$ as in Eq. (10). This is the familiar tridiagonal moment construction and additive resistance structure [1, 2]. No expansion in μ, η, or γ_0 is used here.

The covariance does converge at each finite N. Let $\mathcal{T}$ be the homogeneous part of Eq. (9). Its Hilbert–Schmidt dissipation form is

$$\mathrm{Re}\langle X, \mathcal{T}X\rangle = -\tfrac{1}{2}\sum_{jk}(k_j + k_k)|X_{jk}|^2 - \sum_{j\neq k}(\gamma_j + \gamma_k)|X_{jk}|^2 \leq 0. \tag{12}$$

If an eigenmatrix had an eigenvalue with zero real part, its boundary rows and columns would vanish. Its first-row equation would then force the second row to vanish because $v > 0$. Repeating this argument along the open path forces the entire matrix to vanish. Every eigenvalue of $\mathcal{T}$ therefore has negative real part, which proves the finite-dimensional covariance convergence. Positivity of the finite spin evolution and the canonical anticommutation relations give $0 \leq G_N \leq \mathbb{I}$. This argument makes no claim about a size-independent decay rate of the many-body state.

Gauge the covariance by $U_{jj} = \mathrm{i}^{j-1}$ and index sites by $m = j - L - 1$. Under Eq. (5), Eqs. (10) and (11) become

$$G'_N = U^\dagger G_N U = \tfrac{1}{2}\mathbb{I} + \frac{B_N}{R_N}, \qquad R_N = \frac{\mathcal{R}}{J} = 18 + 2(N-1)q_N, \tag{13}$$

where B_N is a real symmetric tridiagonal matrix with

$$(B_N)_{m,m+1} = 1, \qquad (B_N)_{mm} = q_N m + \tfrac{5}{2}\,\mathrm{sgn}(m) + \mathrm{sgn}(m)\delta_{|m|,L}. \tag{14}$$

Both B_N and R_N are dimensionless. The linear diagonal term is a representation of the stationary density gradient. No magnetic field has been added to the Hamiltonian.

4 Occupation spacing and its finite-size bound

Let $\lambda_+(N)$ be the smallest positive eigenvalue of B_N. Reflection combined with alternating site signs anticommutes with B_N. An irreducible real Jacobi matrix has simple eigenvalues: its first component fixes an eigenvector by recurrence and cannot vanish. The odd-dimensional spectrum consequently consists of L pairs of opposite eigenvalues and one simple zero. Thus G_N has a unique occupation at $1/2$. Define

$$\delta\nu_N = \min_{\nu_a \neq 1/2} |\nu_a - \tfrac{1}{2}| = \frac{\lambda_+(N)}{R_N}. \tag{15}$$

Proposition 1. *For the model and parameters in Eqs. (1)–(6), each fixed finite $c > 0$ has*

$$\lim_{\substack{N\to\infty \\ N \text{ odd}}} \delta\nu_N = \frac{1}{36 + 4c}. \tag{16}$$

For every integer $2 \le M < L$, the finite-chain enclosure is

$$\tfrac{1}{2} + q_N \;\le\; \lambda_+(N) \;\le\; \tfrac{5}{2} + q_N M - 2\cos\!\left(\frac{2\pi}{M+1}\right). \tag{17}$$

The lower bound also holds when no such M exists.

Proof. Delete the central row and column of B_N. The right block T has diagonal $5/2 + q_N j + \delta_{j,L}$, $1 \le j \le L$, and unit off-diagonal entries. The left block is unitarily equivalent to $-T$. Write the ordered eigenvalues of T as $\beta_1 < \cdots < \beta_L$. The path adjacency has operator norm at most 2, so $\beta_1 \ge 1/2 + q_N > 0$. Cauchy interlacing, with the exact central zero counted, gives

$$\beta_1 \le \lambda_+(N) \le \beta_2 \qquad (L \ge 2). \tag{18}$$

For $L = 1$ interlacing still gives the lower bound.

To bound β_2, take the two lowest sine eigenmodes of the length-M unit-hopping path and extend them by zero to the rest of T. This two-dimensional trial space avoids the endpoint shift because $M < L$. On its support the diagonal is at most $5/2 + q_N M$; the largest adjacency Rayleigh quotient within the space is $-2\cos[2\pi/(M+1)]$. The second min–max principle proves the upper bound in Eq. (17). A single trial vector would not suffice for this second-eigenvalue estimate.

For sufficiently large N, choose $M = \lfloor N^{1/3} \rfloor$. It is admissible, and

$$0 \le \lambda_+(N) - \tfrac{1}{2} \le q_N M + \frac{4\pi^2}{(M+1)^2} \longrightarrow 0. \tag{19}$$

Since $R_N \to 18 + 2c$, Eq. (16) follows. □

The upper excess is $O(N^{-2/3})$ for fixed c; the constants are not asserted to be uniform in unbounded c. The trial-space argument is needed because $q_N m$ remains of order one near the remote ends, despite vanishing at every fixed m.

5 Convergence of the normalized orbital

Let u_N be the normalized zero eigenvector of B_N and define $\mathrm{IPR}(u) = \sum_m |u_m|^4$. The physical natural orbital is Uu_N, which has the same IPR. Set $z = 1/2$ and introduce a normalized finite truncation of the zero-background profile:

$$w_0 = C_L, \qquad w_{-m} = C_L z^m, \qquad w_m = C_L(-z)^m \quad (1 \le m \le L), \tag{20}$$
$$C_L^{-2} = 1 + 2\sum_{m=1}^{L} z^{2m}.$$

The infinite profile solves the bulk step recurrence because $z + z^{-1} = 5/2$. Its finite-chain residual has a background part $q_N m w_m$ and endpoint terms of magnitude $(1+z)C_L z^L$. Hence

$$\begin{aligned}\epsilon_L := \|B_N w\|_2 &\le q_N \left(\sum_m m^2 |w_m|^2\right)^{1/2} + \sqrt{2}(1+z)C_L z^L \\ &\le q_N \sqrt{8/9} + \sqrt{2}(1+z)C_L z^L.\end{aligned} \tag{21}$$

The second moment of the infinite normalized geometric distribution is $8/9$. Truncation conditions that distribution on $|m| \le L$, which can only decrease its second moment.

All nonzero eigenvalues of B_N have magnitude at least $g_N = 1/2 + q_N$. Spectral decomposition therefore gives

$$\|(\mathbb{I} - u_N u_N^\dagger) w\|_2 \le \epsilon_L / g_N, \qquad \|u_N - w\|_2 \le \sqrt{2}\,\epsilon_L / g_N, \tag{22}$$

where the phase of u_N is chosen to make $\langle u_N, w\rangle \ge 0$. The second inequality follows from $2(1 - \sqrt{1-r^2}) \le 2r^2$ for the norm r of the perpendicular component. On the unit ball, the Euclidean norm of the gradient of $\sum_m |x_m|^4$ is at most 4. Thus

$$|\,\mathrm{IPR}(u_N) - \mathrm{IPR}(w)| \le \frac{4\sqrt{2}\,\epsilon_L}{1/2 + q_N}. \tag{23}$$

This bound is valid even when its right-hand side is larger than one.

For either scaling in Eq. (6), the right-hand side vanishes as $O(N^{-1})$ plus an exponentially small boundary term. The truncated vector converges in ℓ^2, after embedding about the center, to the normalized infinite geometric vector. Summing its fourth powers gives the second limiting result,

$$\lim_{\substack{N\to\infty \\ N \text{ odd}}} \mathrm{IPR}(u_N) = \frac{(1-z^2)(1+z^4)}{(1+z^2)^3} = \frac{51}{125}. \tag{24}$$

Its central probability is $3/5$ and the probabilities decrease by a factor $1/4$ per site away from the center. In the ungauged basis the amplitudes are proportional to $(-\mathrm{i}/2)^{|m|}$, up to an overall phase. This is the known imaginary-impurity exponential profile [5]; identifying it here does not identify the covariance spectrum with a non-Hermitian energy spectrum.

6 Finite-size illustration and physical consequence

Figure 1 shows the approach to both limits for $q_N = c/N$ at $c = 1/2$ and $c = 2$. It uses $N = 81, 161, 321, 641, 1281, 2561$. The numerical eigenpairs come from a real symmetric tridiagonal eigensolver, retaining the exact reflection pattern when assembling the diagonal. The plotted upper occupation bound minimizes Eq. (17) over all admissible integers M. The IPR bound evaluates Eq. (23) and adds $|\,\mathrm{IPR}(w) - 51/125|$.

The calculation includes both normalizations in Eq. (6), giving 24 cases. At the four largest-size cases, an 80-digit boundary recurrence checks the zero-orbital IPR, and an alternate tridiagonal QR method checks the spacing. The accompanying code records the residuals and discrepancies. These are numerical precision checks; Eqs. (19) and (23) establish the limits independently of the table.

Table 1 also shows that the occupation-spacing limit is approached more slowly than the IPR limit. Even at $N = 2561$, the spacing differs from its limit by about 1.6% for $c = 1/2$ and 4.1% for $c = 2$. The two finite-size normalizations give slightly different values while sharing both limits.

The joint result separates the effects of distributed noise on an orbital and on its occupation environment. For the specified defect, the local orbital shape converges to the zero-background shape, but the denominator R_N retains the finite background contribution $2c$. The occupation separation is consequently reduced to $(36 + 4c)^{-1}$ while the IPR remains $51/125$.

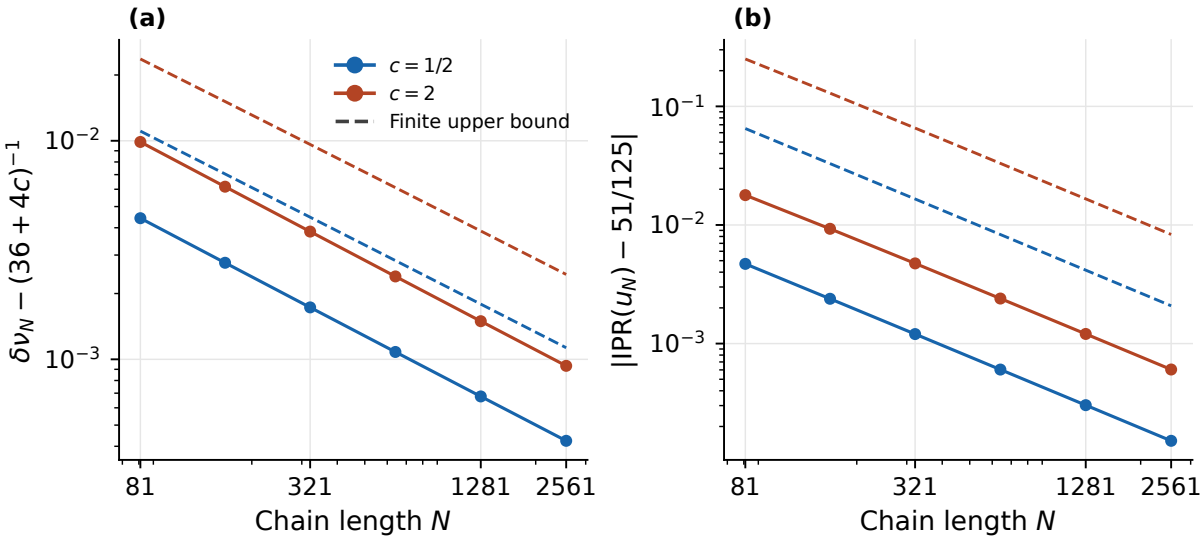


Figure 1: Finite-size deviations from the stationary-first limits at $\kappa/J = 4$, $\mu = 1/2$, $\eta/J = 5$, and $q_N = c/N$. (a) Occupation-spacing excess. (b) Absolute IPR error. Solid lines join numerical values, and dashed lines evaluate the finite upper bounds in Eqs. (17) and (23). Both axes in each panel are logarithmic. The bounds are not fitted to the data. Floating-point evaluations illustrate the analytic inequalities; they are not interval enclosures.

Table 1: Finite values at $N = 2561$ for both background conventions. All quantities displayed are dimensionless, and numerical entries are rounded to nine decimal places. The limiting IPR is 0.408 in every row. The difference between a finite value and its limit is a size effect, not an estimated numerical uncertainty.

c	q_N	$\delta\nu_N$	$(36 + 4c)^{-1}$	$\mathrm{IPR}(u_N)$
1/2	$c/(N-1)$	0.026739105	0.026315789	0.408151461
2	$c/(N-1)$	0.023659581	0.022727273	0.408604709
1/2	c/N	0.026739543	0.026315789	0.408151401
2	c/N	0.023661015	0.022727273	0.408604473

At fixed positive noise per site, the conclusion about the spacing changes. Keeping $q_N = q > 0$ and taking $M = 2$ in Eq. (17) gives, for $L > 2$,

$$\frac{1/2 + q}{18 + 2(N-1)q} \leq \delta\nu_N \leq \frac{7/2 + 2q}{18 + 2(N-1)q}. \tag{25}$$

Thus the spacing has order N^{-1} in that limit. The nonzero limit in Eq. (16) is specifically a finite-total-rate statement, rather than stability of the occupation separation under fixed positive dephasing at every site.

7 Conclusion

For an odd open XX chain with a centered defect and the fixed rates in Eq. (5), finite distributed phase-flip rate preserves an isolated half-occupied natural orbital. Its spacing tends to $(36+4c)^{-1}$ while its normalized IPR tends to 51/125. Interlacing and a two-mode variational estimate control the spectral edge despite the nonvanishing diagonal variation far from the center. A residual estimate separately controls the orbital norm and IPR. These provide finite-size benchmarks for stationary one-particle calculations in this specified noise model.

The result concerns neither a Liouvillian gap nor an excitation energy, and no Gaussian-state assumption is made. It does not establish a reversed order of the time and size limits, uniformity as c grows without bound, an off-center or variable-strength defect result, or an extension to anisotropic XY chains. The broader classification of spatial dissipation profiles remains outside the result proved here.

Data and code availability

The finite-chain data, a Python reproduction program, environment information, and instructions for regenerating the figure and table accompany this manuscript. All plotted data are calculations in the specified finite spin model; no experimental data are used.

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A Normalization of the uniform-chain reference

The dissipator and ladder definitions of Ref. [1], arXiv version 1, map to those used here by $\kappa = 4\Gamma$ and $\gamma_j = 2\gamma_{\mathrm{ref}}$ when $J = 1$. For $N = 4$, $\Gamma = \gamma_{\mathrm{ref}} = 1$, $\mu = 1/2$, and no defect, Eq. (11) gives

$$(\langle\sigma_1^z\rangle, \ldots, \langle\sigma_4^z\rangle) = (-2/5, -1/10, 1/10, 2/5), \qquad b = -1/10, \tag{26}$$

in agreement with Eqs. (13)–(14) of that reference. There is a current-normalization qualification in the cited preprint: the literal operator $j^z = 2(\sigma_j^x\sigma_{j+1}^y - \sigma_j^y\sigma_{j+1}^x)$ has expectation $4b = -2/5$, whereas the prose following Eq. (14) gives $2b$. The latter is the current of the occupation $(1 + \sigma^z)/2$, equal to half the magnetization current. All equations here use the explicitly defined spin operators and canonical dissipator, so the occupation result is independent of that naming ambiguity.

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